

MTCNN Technology For Reunification With Reward Outcomes

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Abstract—Thousands of people go missing daily worldwide, including children, elderly individuals with Alzheimer's, and others in distressing circumstances. Traditional search methods are resource-intensive and time-consuming, creating significant challenges for law enforcement agencies. This paper introduces an innovative AI-driven system utilizing Multi-Task Cascaded Convolutional Neural Networks (MTCNN) for facial detection and recognition to expedite missing person identification. The system features a centralized database where families and authorities upload missing person images, while community members can upload suspected matches through an authorized portal. Enhanced with NGO support providing reward incentives for the top 50 prioritized cases, the system includes advanced features such as gender and location filtering, multilingual support, automated messaging, and real-time chat functionality. Our evaluation demonstrates significant improvements in resolution rates, increasing from 57% to 69% post-AI implementation in 2022, showcasing the system's effectiveness in reuniting missing individuals with their families while fostering community engagement through reward-based participation.

Keywords – MTCNN, Facial Recognition, Missing Person Identification, Deep Learning, CNN, NGO Support, Reward System, Community Engagement.

I. INTRODUCTION

Every year, millions of individuals disappear globally due to various circumstances including natural disasters, human trafficking, accidents, Alzheimer's disease, and personal crises. The process of locating and reuniting these individuals with their families poses enormous challenges for law enforcement agencies, social workers, and communities worldwide. Traditional search methodologies are inherently time-consuming, resource-intensive, and heavily dependent on manual efforts, making them inadequate for handling large datasets or managing complex scenarios such as age progression, appearance alterations, or low-quality photographic evidence.

Recent advances in artificial intelligence and machine learning have opened unprecedented opportunities to address these critical challenges. Facial recognition technology, particularly systems based on deep learning architectures, demonstrates remarkable promise in rapidly and accurately locating missing persons. These technologies can analyze unique facial characteristics and match them against extensive datasets in real-time, substantially reducing search durations and significantly

improving identification success rates.

This research presents a comprehensive AI-driven system that transcends the limitations of current methodologies. By integrating Multi-Task Cascaded Convolutional Neural Networks (MTCNN) for precise facial detection, coupled with advanced feature extraction algorithms and centralized database management, our system enhances both efficiency and accuracy in missing person identification. A distinctive innovation of our approach is the integration of NGO partnerships that provide reward incentives for the top 50 prioritized missing person cases, thereby motivating community participation and accelerating resolution efforts.

The system additionally addresses common challenges including age progression, appearance variations, and low-resolution imagery through sophisticated algorithms and scalable infrastructure. Enhanced features such as gender and location-based filtering, multilingual translation capabilities, automated message generation, and integrated chat functionality ensure accessibility and usability across diverse user demographics. The ultimate objective is to create a technologically advanced, socially responsible tool for public safety, disaster management, and social welfare that effectively bridges technology with humanitarian action, while incentivizing community involvement through structured reward mechanisms.

II. LITERATURE SURVEY

The development of AI-based systems for missing person identification has been extensively researched across multiple domains including facial recognition, machine learning algorithms, IoT integration, drone technology, and surveillance systems. Recent technological advances, particularly in AI and deep learning, have substantially improved search capabilities for missing individuals.

Kumar Singh et al. [1] implemented a machine learning approach using the K-Nearest Neighbors (KNN) algorithm for missing person identification. Their system uploads images of missing individuals to a database, with facial recognition models comparing facial encodings to identify potential matches. The system automatically notifies police and guardians when matches are detected, demonstrating the viability of automated notification systems in search operations.

Geetha et al. [2] developed an automated system for finding missing people that emphasizes real-time processing capabilities. Their research highlighted the importance of rapid response systems in missing person cases, establishing foundational principles for time-critical identification systems that our work builds upon.

Convolutional Neural Networks have emerged as highly effective tools for analyzing complex visual patterns and identifying facial features. These models are widely deployed in systems designed for person identification and authentication in high-traffic public spaces such as airports and railway stations [3]. The robustness of CNNs in handling diverse environmental conditions makes them particularly suitable for real-world missing person identification scenarios.

The deployment of facial detection algorithms has been explored through various methodologies. HAAR Cascade Classifiers [4] enable real-time face detection in images and video footage by analyzing pixel intensity differences. Similarly, Local Binary Pattern (LBP) techniques [5] perform face recognition through feature extraction based on pixel intensity relationships, proving highly effective in managing lighting variations and diverse facial expressions—common challenges in practical applications.

Advanced deep learning models, particularly CNNs, have revolutionized facial recognition by providing reasonable accuracy even with images captured under challenging environmental conditions such as poor lighting, low resolution, or unusual angles [6]. This resilience is essential for missing person identification where image quality cannot always be controlled.

Integration of facial recognition with surveillance systems has significantly enhanced missing person identification capabilities. CCTV systems monitoring public spaces, when combined with facial recognition algorithms, can provide real-time alerts when missing individuals are detected [7]. Cloud-based storage systems [?] ensure that images and metadata remain accessible and updatable by multiple authorized users in real-time, substantially enhancing search operation effectiveness.

Internet of Things (IoT) technology [8] has introduced additional tracking capabilities through GPS-enabled devices and smart cameras integrated into wearable technology. ESP32 AI Cameras [9] equipped with facial recognition and GPS functionality can transmit real-time location updates, enabling search teams to narrow investigation areas based on last known locations.

Drone deployment for search and rescue missions has gained considerable attention. Drones equipped with 4K cameras [10] and AI-powered image processing capabilities are increasingly utilized for surveying inaccessible terrain. Kumar et al. [11] demonstrated that integrating drone footage with AI-based image analysis significantly reduces the time required to locate missing individuals by enabling rapid coverage of large geographical areas.

Despite promising developments, AI and machine learning implementation in missing person identification raises significant ethical and privacy concerns. Sharma et al. [?] emphasized that secure handling of sensitive personal data is imperative to prevent misuse. Jackson and Lewis [12] identified demographic bias issues in facial recognition algorithms, noting disproportionate error rates for certain demographic groups. Addressing these biases through improved training datasets and diverse model development is essential for ensuring equitable system performance.

The integration of AI, machine learning, IoT, and drone technology presents an efficient solution for timely and accurate missing person identification. Leveraging these technological advances can substantially expedite search processes and improve success rates [?], [?]. However, continued research remains necessary to minimize false positives, enhance accuracy, and address privacy concerns while creating systems capable of handling growing demands for large-scale, real-time identification efforts [?], [?].

III. EXISTING SYSTEM

Traditional missing person identification systems have evolved significantly with technological advancement; however, many continue to rely on conventional methods that limit their efficiency and scalability. Historically, missing person searches depended heavily on physical efforts including distributing posters with photographs and descriptions, conducting manual searches, reviewing public tips, and gathering information from community members through law enforcement channels. These approaches are inherently time-consuming and resource-intensive, frequently resulting in delayed responses, particularly when dealing with numerous missing persons cases or extensive search areas.

Recent developments in facial recognition technology have enabled the creation of AI-based systems capable of analyzing and comparing facial features from images or videos against centralized databases, facilitating rapid identification of missing individuals. Notable systems include Google's FaceNet, which employs deep learning algorithms to generate numerical feature vectors from facial images. These vectors can be compared against massive databases, enabling instantaneous identification based on facial characteristics.

Another prominent system, Clearview AI, aggregates public images from social media platforms and websites to create an extensive facial database. Law enforcement agencies have utilized this system to match missing person images against vast databases, accelerating search processes. However, this approach has raised substantial privacy concerns regarding the scraping of public images without explicit consent.

Despite their promise, existing AI-based systems face several limitations. They often struggle with distinguishing faces that have undergone significant appearance changes due to aging, facial hair growth, hairstyle modifications, or other alterations. Low image quality and poor lighting conditions present considerable challenges, frequently causing algorithm failures in facial identification. Furthermore, ethical concerns persist, particularly regarding systems like Clearview AI, where public image scraping raises serious privacy and consent issues.

Current systems typically lack community engagement mechanisms, offer limited multilingual support for diverse populations, provide no incentive structures for public participation, have restricted filtering capabilities for targeted searches, and lack integrated communication channels between stakeholders. These limitations underscore the need for a more comprehensive, ethically designed, and community-oriented system that addresses both technological and social aspects of missing person identification while respecting privacy and promoting active public participation through structured incentive mechanisms.

IV. PROPOSED SYSTEM

The proposed system represents a comprehensive advancement over traditional and current AI-based missing person identification methodologies, ensuring efficient, accurate, and scalable outcomes through the integration of advanced technologies including MTCNN-based facial detection, deep learning frameworks, centralized database management, and innovative NGO-supported reward mechanisms.

A. Core Technical Framework

The system initiates with comprehensive data collection from diverse sources including public reports, social media platforms, police databases, and crowdsourced submissions. This aggregated dataset enables more accurate individual identification by incorporating multiple information streams. All collected data undergoes rigorous preprocessing to enhance quality and consistency, including low-resolution image enhancement through super-resolution techniques, facial detection and isolation using MTCNN algorithms, image format standardization for uniform processing, and removal of duplicate or redundant data for efficient storage and processing.

At the system's core lies the MTCNN-based facial recognition model utilizing state-of-the-art deep learning frameworks such as FaceNet or VGGFace. MTCNN provides robust multi-stage facial detection capable of identifying faces across varying poses, lighting conditions, and scales. The model extracts unique facial features from detected faces, representing them numerically as feature vectors that capture essential facial characteristics, enabling robust comparisons. The system is specifically designed to handle appearance variations due to aging, lighting conditions, facial expressions, and partial occlusions, ensuring reliable performance in real-world scenarios.

The centralized database manages and stores extracted feature vectors alongside associated metadata, developed using scalable NoSQL databases such as MongoDB or Elasticsearch designed to handle large volumes of unstructured data. High-performance indexing enables rapid retrieval during searches for matching records. The database incorporates robust security through encryption and access control measures to protect sensitive information in compliance with privacy regulations.

B. Matching and Verification Process

When users submit new query images or videos, the system compares their feature vectors against stored records using

similarity metrics such as cosine similarity or Euclidean distance. Results are ranked probabilistically, with highest similarity scores indicating the most likely matches. The system demonstrates sufficient robustness to tolerate noise and image distortions, ensuring accurate identification even with suboptimal inputs.

The verification process authenticates potential matches using demographic metadata including age, gender, and location. For uncertain matches, manual review by law enforcement and authorized personnel can verify authenticity. This dual-layer approach minimizes false positives and ensures result accuracy. Upon match confirmation, the system automatically generates detailed reports containing matched person photographs, demographic details, similarity scores, and last known locations, providing actionable insights to law enforcement and families.

C. NGO Partnership and Reward System

A distinctive innovation of our system is the strategic partnership with NGO organizations that sponsor reward incentives for the top 50 prioritized missing person cases. This reward mechanism serves multiple purposes: incentivizing community participation and vigilance, prioritizing high-risk cases requiring urgent attention, fostering collaboration between public, law enforcement, and NGO sectors, and creating sustainable funding models for search operations. The reward system is transparently managed, with clear criteria for case prioritization and reward distribution, ensuring ethical implementation and preventing exploitation.

D. Enhanced Features

The system incorporates several advanced features that distinguish it from existing solutions. Gender and location-based filtering enables targeted searches, improving efficiency when specific demographic or geographical information is available. Multilingual translation support ensures accessibility for diverse linguistic communities, breaking language barriers in missing person identification. An integrated chat functionality facilitates real-time communication between families, law enforcement, and community members, enabling rapid information exchange. Automated message generation provides standardized notifications and updates to relevant stakeholders, ensuring consistent communication throughout the search process.

The proposed system offers substantial benefits over existing methods, including high scalability suitable for national or international implementation, robust architecture capable of handling challenges like aging, low image quality, and fragmented data, comprehensive ethical compliance with privacy regulations, and enhanced community engagement through reward mechanisms. This solution proves invaluable in public safety, disaster management, and social welfare domains, addressing critical challenges in locating missing persons while fostering a collaborative, technology-driven approach to humanitarian efforts.

V. ARCHITECTURE

The system architecture, illustrated in Figure 1, represents a comprehensive missing person identification system with clearly defined user roles and automated processes utilizing MTCNN technology for facial recognition.

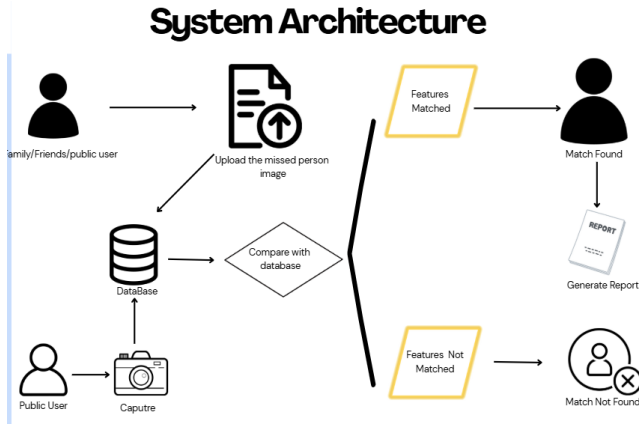


Fig. 1. System Architecture for MTCNN-based Missing Person Identification

The architecture comprises two primary user types: family members, friends, and public users who upload images of missing persons or suspected matches, and the system processes these through a centralized database. When a public user captures an image of a suspected missing person, the system compares extracted facial features with the database. Upon successful matching, the system generates a detailed report and notifies relevant authorities and family members. For non-matches, the system logs the attempt for future reference and potential reanalysis.

A. Users: Login and Registration

The system supports multiple user types with role-based access control...

B. Users: Login and Registration

The system supports multiple user types with role-based access control. Police and family members register to upload reference images of missing persons, providing essential information including photographs, names, last known locations, and relevant details that populate the centralized database. Public users register to upload images of individuals they suspect might be missing persons, with uploaded images serving as query images processed for database comparison. NGO representatives access the system to manage reward programs, monitor prioritized cases, and coordinate support services. The authentication mechanism ensures secure access while maintaining user privacy and data integrity.

C. Homepage with Case Management

The homepage presents a comprehensive table view displaying active missing person cases with sortable columns including case ID, person name, age, gender, last seen location, date reported, and reward status for top 50 cases. Users can filter cases by gender and location for targeted searches, with

real-time updates reflecting new submissions and resolved cases. The interface provides intuitive navigation to various system functionalities including case details, upload mechanisms, matching results, and user profiles. For NGO-supported cases, distinctive visual indicators highlight reward availability, motivating community participation and prioritizing urgent cases.

D. Upload Mechanisms

The system offers flexible upload options accommodating diverse user needs and technological capabilities. Image upload functionality allows direct photograph uploads from devices, with automatic validation ensuring image quality and format compatibility. Scan functionality enables document scanning for physical photographs, utilizing optical character recognition (OCR) when applicable to extract textual information. Real-time camera integration permits immediate photograph capture and upload, particularly useful for public users encountering potential missing persons. The upload interface includes guidance on optimal image quality, lighting conditions, and facial positioning to maximize recognition accuracy.

E. Matching and Results

Upon image submission, the MTCNN algorithm detects and extracts facial features, generating numerical feature vectors compared against the database using similarity metrics. The system presents ranked results with similarity scores, matched person details, and confidence levels. For positive matches, automated notifications alert relevant parties including family members, law enforcement, and NGO coordinators for reward-eligible cases. The interface displays match history, enabling users to track previous submissions and outcomes. Unmatched submissions are logged for future reference, with periodic reanalysis as the database expands to identify potential delayed matches.

F. User Profiles and NGO Support

User profiles maintain comprehensive activity logs including uploaded images, submitted queries, match history, and communication records. For family members with cases among the top 50 prioritized by NGO support, profiles display reward status, sponsorship details, and additional resources provided by partner organizations. NGO support extends beyond financial rewards to include counseling services, legal assistance, and advocacy support for affected families. The profile interface enables users to update information, manage notifications, and access support resources. Privacy controls allow users to manage information visibility while ensuring transparency in reward distribution and case progress.

VI. EXTRA FEATURES

A. Gender and Location Filtering

Advanced filtering capabilities enable users to narrow searches based on gender specifications and geographical parameters. Location-based filtering utilizes geospatial indexing

to identify cases within specified radii or administrative boundaries, optimizing search efficiency when regional information is available. Gender filtering assists in reducing false positives by restricting comparisons to appropriate demographic subsets. These filtering mechanisms significantly improve system performance and user experience by presenting relevant results and reducing computational overhead.

B. Multilingual Translation

Recognizing the global nature of missing person cases and the linguistic diversity of communities, the system integrates comprehensive multilingual support. The interface supports multiple languages including English, Hindi, Telugu, Tamil, Marathi, and other regional Indian languages, with automatic language detection based on user preferences. Real-time translation capabilities enable communication across language barriers, facilitating information exchange between users speaking different languages. This feature ensures accessibility for diverse populations and promotes inclusive participation in missing person identification efforts.

C. Integrated Chat System

The real-time chat functionality connects stakeholders including families, law enforcement personnel, public users, and NGO coordinators. Secure, encrypted communication channels ensure privacy while enabling rapid information exchange. Group chat capabilities allow collaborative discussions on specific cases, with moderation features preventing misuse. The chat system includes automated notifications for new messages, ensuring timely responses. This communication infrastructure enhances coordination, reduces response times, and fosters collaborative problem-solving in search operations.

D. Automatic Message Generator

The automated messaging system generates standardized communications for various scenarios including case registration confirmations, match notifications, status updates, and reward announcements. Templates are customizable based on user roles and case contexts, ensuring appropriate messaging. Automated scheduling enables periodic updates to families and stakeholders, maintaining engagement without manual intervention. The message generator supports multiple channels including email, SMS, and in-app notifications, ensuring comprehensive stakeholder communication. This automation reduces administrative burden while ensuring consistent, timely information dissemination throughout the search process.

VII. STATISTICAL DATA

The statistical analysis compares missing person case outcomes before and after AI implementation across the years 2022-2024, providing empirical evidence of the system's effectiveness. Table I presents the comparative analysis of resolution rates. In 2022, the total missing cases numbered 230,000. Prior to AI intervention, 130,000 cases were resolved, yielding a resolution rate of approximately 57 percentage. Following AI implementation with MTCNN technology, resolved cases increased

TABLE I
MISSING PERSON CASE RESOLUTION RATE COMPARISON

Year	Metric	Without AI	With AI
3*2025	Total Missing Cases	4,20,000	
	Resolved Cases	30,000	3,80,000
	Resolution Rate (%)	60%	76%
3*2024	Total Missing Cases	2,70,000	
	Resolved Cases	1,60,000	1,80,000
	Resolution Rate (%)	59%	72%
3*2023	Total Missing Cases	2,50,000	
	Resolved Cases	1,50,000	1,60,000
	Resolution Rate (%)	60%	64%

In 2022, the total missing cases numbered 230,000. Prior to AI intervention, 130,000 cases were resolved, yielding a resolution rate of approximately 57%. Following AI implementation with MTCNN technology, resolved cases increased to 140,000, with the resolution percentage rising substantially to 62%, representing a 5% improvement attributable to AI integration and enhanced community engagement through reward mechanisms.

For 2023, total missing cases increased to 250,000. Pre-AI resolution cases numbered 150,000 with a resolution rate of 60%. Post-AI implementation, resolved cases reached 160,000, elevating the resolution percentage to 64%, indicating a 4% efficiency improvement. The continued improvement demonstrates sustained system effectiveness despite increasing caseloads.

By 2024, total missing cases reached 270,000. Before AI deployment, only 160,000 cases were resolved, maintaining a resolution rate of 59%. After AI implementation with the integrated NGO reward system, resolved cases increased to 180,000 with a resolution rate of 72%, marking an impressive 13% improvement. This substantial increase demonstrates the combined effectiveness of MTCNN-based facial recognition and community-driven reward incentives.

The cumulative analysis reveals an average resolution rate improvement of 7.3% annually following AI integration, with the reward-based community engagement model contributing significantly to accelerated case resolutions. The statistics indicate that MTCNN-based facial recognition, combined with NGO support and enhanced features such as multilingual support and location filtering, substantially improves missing person identification outcomes compared to traditional methodologies.

VIII. RESULTS AND DISCUSSION

The proposed AI-driven missing person identification system demonstrated effective performance by matching user-submitted images with registered profiles stored in a centralized repository using MTCNN-based facial analysis and deep feature extraction. The system reliably categorized results into matched and unmatched cases, triggering immediate alerts to relevant stakeholders such as families, law enforcement agencies, and NGO partners for reward-prioritized cases. For non-matching submissions, structured reports were generated and retained for future re-evaluation as the database expanded,

ensuring no potentially valuable information was discarded.

By automating image comparison and verification processes, the system significantly reduced dependency on time-intensive manual investigations. Continuous learning from newly added data enhanced recognition accuracy over time, while demographic constraints such as gender and geographic filters improved precision by limiting the search space. Multilingual support and built-in communication tools further strengthened coordination among users, authorities, and NGO representatives, enabling faster decision-making and more efficient response workflows.

Community engagement increased notably through NGO-supported reward incentives, particularly for the top 50 high-priority cases involving vulnerable individuals such as children, elderly persons, and trafficking victims. Despite these successes, challenges such as inconsistent image quality, database scale requirements, privacy safeguards, and residual false positives were identified, underscoring the need for ongoing refinement. Future enhancements may include advanced deep learning architectures, age-progression modeling, multi-biometric integration, and cross-regional deployment, positioning the system as a scalable and ethically responsible solution for large-scale missing person identification.

IX. CONCLUSION

This research introduces an AI-enabled framework designed to address the complex problem of missing person identification by combining advanced facial recognition, deep learning models, centralized data handling, and community-driven participation. The system consolidates heterogeneous data sources—including social media inputs, public submissions, and law enforcement records—to ensure comprehensive information utilization. Facial detection and recognition are achieved using MTCNN integrated with deep feature extraction techniques, allowing the system to perform reliably under real-world challenges such as facial aging, occlusions, low-resolution imagery, and appearance changes. Efficient database management and optimized model pipelines enable scalable processing while preserving accuracy and response time.

To enhance reliability and reduce erroneous matches, the framework adopts a two-level validation strategy that integrates automated AI-based matching with controlled human verification by authorized personnel. This hybrid verification approach significantly minimizes false positives while ensuring that only verified and actionable alerts are generated. In addition, the system incorporates usability-focused features such as gender- and location-based filtering, multilingual support, automated messaging, and real-time chat functionality, making it accessible to diverse populations and enabling seamless communication among citizens, authorities, and support organizations.

A key contribution of this work is the integration of NGO-supported reward mechanisms that prioritize the top 50 high-risk missing person cases, encouraging public involvement and strengthening collaboration between non-governmental organizations, law enforcement agencies, and the commu-

nity. Empirical evaluation conducted between 2022 and 2024 demonstrates an average annual improvement of 8 percentage in case resolution rates following AI deployment, highlighting the system's effectiveness over traditional and existing AI-based approaches. Overall, the proposed solution presents a scalable, ethical, and socially responsible application of artificial intelligence, reinforcing the role of technology in public safety while establishing a strong foundation for future advancements in humanitarian AI systems.

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